

International Institute of Information Technology, Bangalore
CSE 511 Algorithms: Practice Problem

Description: Given the lengths of several ropes, you must connect all the ropes into one single rope. The cost of connecting two ropes is equal to the sum of their lengths. Return the minimum total cost required to connect all the ropes.

Input: The first line contains an integer n , the number of ropes. The second line contains n integers representing the lengths of the ropes.

Output: Print the minimum total cost to connect all ropes into one.

Example:

Input :

4

4 3 2 6

Output :

29

Explanation:

- Connect ropes of length 2 and 3: cost = 5. Ropes become [4, 5, 6].
- Connect ropes of length 4 and 5: cost = 9. Ropes become [6, 9].
- Connect ropes of length 6 and 9: cost = 15. Ropes become [15].
- Total cost = $5 + 9 + 15 = 29$.

Constraints:

- $1 \leq n \leq 10^5$
- $1 \leq \text{ropeLength}[i] \leq 10^4$